

A Little Book About Fractions

Introduction

This is what a fraction looks like:

$$\frac{x}{y}$$

Here is an example:

$$\frac{1}{2}$$

"Fraction" really just means "divide these two numbers." Fractions make things easier when the answer to the division problem isn't a whole number, like 7 divided by 4, 13 divided by 20...or 1 divided by 2.

The Basics

Rule 1. Fractions have many names.

Basically, these all mean the same thing:

$\frac{x}{y}$
 $x \div y$
 x/y
 x yths
 x divided by y
the quotient of x and y
 x out of y
what is the proportion of x to y
how many y s are in x

Example:

12 divided by 4
= the quotient of 12 and 4
= how many 4s are in 12
= twelve fourths
= 3

Example:

$$\frac{20}{5}$$
$$= 20/5$$
$$= 4$$

Example:

$$1792 \div 56$$
$$= \text{what is the proportion of 1792 to 56}$$
$$= 32$$

Here are two numbers that we can't divide evenly, so we keep them in fraction form:

$$3 \text{ out of } 10$$
$$= 3/10$$
$$= \frac{3}{10}$$

Rule 2. 1 in the "denominator"

Any number divided by 1 is the number itself.

$$\frac{x}{1} = x$$

Example:

$$\frac{3}{1} = 3$$

Rule 3. 0 in the "numerator"

0 out of any number is always 0.

$$\frac{0}{x} = 0$$

Example:

$$\frac{0}{4} = 0$$

Rule 4. 0 in the "denominator"

You cannot divide any number by 0. Write "undefined" instead.

$$\frac{x}{0} = \text{undefined}$$

Example:

$$\frac{2}{0} = \text{undefined}$$

Rule 5. the "identity property"

A number over itself is always 1.

$$\frac{x}{x} = 1$$

Example:

$$\frac{5}{5} = 1$$

Rule 6. the "reciprocal"

The reciprocal of a fraction is just the fraction flipped upside down, so the denominator and numerator switch places.

$$\frac{x}{y} => \frac{y}{x}$$

Example:

The reciprocal of $\frac{2}{5}$ is $\frac{5}{2}$

Reducing

Reducing means making the top and bottom numbers of the fraction as small as possible without changing the fraction's amount.

For example, two quarters of a cheesecake is the same amount as half the cheesecake.

In other words, reducing $\frac{2}{4}$ gives you $\frac{1}{2}$.

Rule 1. Factors

Factors of a number are the whole numbers you can multiply by each other to get that number. You can say that factors "go into" that larger number or that the larger number is "divisible" by those factors.

A number is always one of its own factors.

For example:

*Multiply 2 and 3 to get 6.
2 and 3 are each a factor of 6.*

Example:

The factors 1, 2, 4, 5, 10, and 20 all go into 20.

Example:

32 is divisible by 1, 2, 4, 8, 16, and 32.

Example:

*All the factors of 100:
1, 2, 4, 5, 10, 20, 25, 50, and 100*

Rule 2. Prime Numbers

Prime numbers are numbers that have exactly **two** different positive factors: 1 and the number itself.

Example: 5

*The factors of 5 are 1 and 5.
That's all of 5's possible factors, so 5 is prime.*

Example: 17

17 is prime because its factors are just 1 and 17.

Example: 8

8 has the four factors 1, 2, 4, and 8, so it is NOT prime

Example: 1

1 only has one factor, 1, so it is NOT prime

Example: 2

2 is prime because its factors are 1 and 2.

Example: 0

*0 is NOT prime because it has infinite factors:
Anything times 0 is 0, so all numbers are its factors.*

Rule 3. Smallest Prime Numbers

These are all the prime numbers less than 100:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43,
47, 53, 59, 61, 67, 71, 73, 79, 83, 89, and 97

Rule 4. Prime Factorization

Prime Factorization gets you the prime numbers you would need to multiply by each other to get a number. You can use more than one of the same prime number to get there.

Here are the steps:

1. Divide your number by the smallest prime number (2) as many times as possible. Stop when dividing wouldn't give a whole number.
2. Take the number you are left with and divide it by the next lowest prime (3) as many times as possible.
3. Continue with each of the primes (5, 7, etc.) until you are left with 1.
4. List how many of each prime you were able to divide by.

Example: find the prime factorization of 924.

(2)

With 924, try dividing it by 2

$$924 \div 2 = 462$$

With 462, try dividing by 2 again

$$462 \div 2 = 231$$

231 is not divisible by 2 again.

(3)

With 231, try dividing it by 3

$$231 \div 3 = 77$$

77 is not divisible by 3 again.

(5)

77 is not divisible by 5.

(7)

With 77, try dividing it by 7

$$77 \div 7 = 11$$

11 is not divisible by 7 again.

(11)

With 11, try dividing it by 11

$$11 \div 11 = 1$$

We got to 1, so we are done.

We divided by 2 twice, 3 once, 7 once and 11 once to get
So the prime factorization of 924 is 2, 2, 3, 7, and 11.

Example: find the prime factorization of 110

(2)

$$110 \div 2 = 55$$

55 is not divisible by 2

(3)

55 is not divisible by 3

(5)

$$55 \div 5 = 11$$

11 is not divisible by 5

(7)

11 is not divisible by 7

(11)

$$11 \div 11 = 1.$$

The prime factorization of 110 is $2 \times 5 \times 11$.

Example: 18

(2)

$$18 \div 2 = 9$$

(3)

$$9 \div 3 = 3$$

$$3 \div 3 = 1$$

So the prime factorization of 18 is $2 \times 3 \times 3$

Example: 8

(2)

$$8 \div 2 = 4$$

$$4 \div 2 = 2$$

$$2 \div 2 = 1$$

So the prime factorization of 8 is 2, 2, and 2

Example: 10

$$10 \div 2 = 5$$

The prime factors of 10 are 2 and 5.

Example: 40

The prime factorization of 40 is $2 \times 2 \times 2 \times 5$.

Example: 39

$$39 = 3 \times 13$$

If you are more comfortable, you can go backwards and start with the highest numbers that would divide evenly into your number, then break those into primes.

Example: 100

$$100 = 10 \times 10$$

$$100 = 2 \times 5 \times 2 \times 5$$

The prime factorization of 100 is 2, 2, 5, and 5.

Rule 5. Reduction

To reduce a fraction, find the prime factorization of the numerator and denominator. Then remove any shared prime factors.

Example:

$$\text{Reduce } \frac{4}{10}$$

The prime factors of 4 are 2 and 2

The prime factors of 10 are 2 and 5

They both share a 2, so remove 2 from both lists.

4 is left with 2

10 is left with 5

Reducing $\frac{4}{10}$ gives you $\frac{2}{5}$

Example:

$$\text{Reduce } \frac{54}{36}$$

$$54 = 2 \times 3 \times 3 \times 3$$

$$36 = 2 \times 2 \times 3 \times 3$$

Remove 2, 3, 3 from both

$$3 \times (2 \times 3 \times 3) \Rightarrow 3$$

$$2 \times (2 \times 3 \times 3) \Rightarrow 2$$

$$\frac{54}{36} = \frac{3}{2}$$

Example: 1 in the numerator

$$\text{Reduce } \frac{1}{3}$$

1 cannot be reduced, so we say this is already reduced

If you are more comfortable, you can find the largest non-prime number that is shared by both and divide your numbers by that to reduce your fraction.

Example:

$$\frac{30}{50}$$

$$30 \div 10 = 3$$

$$50 \div 10 = 5$$

$$\frac{30}{50} = \frac{3}{5}$$

Example:

$$\frac{56}{8}$$

$$= \frac{7 \times (8)}{1 \times (8)}$$

$$= \frac{7}{1}$$

$$\frac{56}{8} = 7$$

Multiplying

Rule 1. No reduction needed

When two fractions are already reduced as far as they can go, just multiply their numerators with each other to get the top number and their denominators with each other to get the bottom number.

$$\frac{x}{y} \times \frac{z}{w} = \frac{x \times z}{y \times w} = \frac{xz}{yw}$$

Example:

$$\frac{4}{5} \times \frac{3}{7} = \frac{4 \times 3}{5 \times 7} = \frac{12}{35}$$

Example:

$$\frac{2}{3} \times \frac{1}{3} = \frac{2 \times 1}{3 \times 3} = \frac{2}{9}$$

If there is a whole number, just stick a 1 under it and proceed as usual.

Example:

$$5 \times \frac{3}{4} = \frac{5}{1} \times \frac{3}{4} = \frac{5 \times 3}{1 \times 4} = \frac{15}{4}$$

Example:

$$\frac{1}{3} \times 3 = \frac{1 \times 3}{3 \times 1} = \frac{1}{1} = 1$$

Rule 2. Reduction needed

When fractions are not already reduced as far as they can go, reduce each of them first and then multiply them as usual.

You can also multiply first and then reduce.

Example:

$$\begin{aligned} & \frac{3}{6} \times \frac{2}{8} \\ &= \frac{1 \times 3}{2 \times 3} \times \frac{1 \times 2}{2 \times 4} \\ & \frac{1}{2} \times \frac{1}{4} = \frac{1}{8} \end{aligned}$$

Example:

$$\frac{2}{6} \times \frac{1}{3} = \frac{2 \times 1}{2 \times 3 \times 3} = \frac{1}{9}$$

Dividing

Rule 1. Multiply by the reciprocal

To divide one fraction by another, just multiply the first fraction by the reciprocal of the second fraction.

$$\frac{x}{y} \div \frac{z}{w} = \frac{x}{y} \times \frac{w}{z}$$

Example:

$$\frac{1}{2} \div \frac{3}{5} = \frac{1}{2} \times \frac{5}{3} = \frac{5}{6}$$

Example:

$$\frac{3}{2} \div \frac{6}{7} = \frac{3}{2} \times \frac{7}{6} = \frac{3 \times 7}{2 \times 2 \times 3} = \frac{7}{2 \times 2} = \frac{7}{4}$$

Example:

$$2 \div \frac{1}{4} = \frac{2}{1} \times \frac{4}{1} = \frac{2 \times 4}{1 \times 1} = \frac{8}{1} = 8$$

Rule 2. Fraction of a fraction

Nested fractions are just division problems:

$$\frac{\frac{x}{y}}{\frac{z}{w}} = \frac{x}{y} \div \frac{z}{w} = \frac{x}{y} \times \frac{w}{z} = \frac{xw}{yz}$$

Example:

$$\frac{\frac{1}{2}}{\frac{3}{4}} = \frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{2 \times 2}{2 \times 3} = \frac{2}{3}$$

Example:

$$\frac{2}{\frac{3}{4}} = 2 \div \frac{3}{4} = \frac{2}{1} \times \frac{4}{3} = \frac{8}{3}$$

Adding

Rule 1. Same denominator

If the denominator is the same, just add the numerators and keep the denominator as is.

$$\frac{x}{y} + \frac{z}{y} = \frac{x+z}{y}$$

Example:

$$\frac{4}{7} + \frac{2}{7} = \frac{4+2}{7} = \frac{6}{7}$$

Example:

$$\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

Remember to reduce if needed.

Example:

$$\frac{6}{5} + \frac{4}{5} = \frac{10}{5} = \frac{5 \times 2}{5} = \frac{2}{1} = 2$$

Rule 2. Different denominator

If the denominators are different, multiply each numerator by the *other* fraction's denominator and add the results to get the new numerator, and multiply the two denominators together to get the new denominator.

$$\frac{x}{y} + \frac{z}{w} = \frac{x \times w + y \times z}{y \times w} = \frac{xw + yz}{yw}$$

Example:

$$\frac{1}{3} + \frac{2}{5} = \frac{1 \times 5 + 2 \times 3}{3 \times 5} = \frac{5+6}{15} = \frac{11}{15}$$

This works because you are really multiplying each fraction by 1:

$$\begin{aligned} \frac{1}{3} + \frac{2}{5} &= \\ &= 1 \times \frac{1}{3} + 1 \times \frac{2}{5} \\ &= \left(\frac{5}{5}\right) \times \frac{1}{3} + \left(\frac{3}{3}\right) \times \frac{2}{5} \\ &= \frac{5 \times 1}{5 \times 3} + \frac{3 \times 2}{3 \times 5} \\ &= \frac{5}{15} + \frac{6}{15} \\ &= \frac{5+6}{15} \\ &= \frac{11}{15} \end{aligned}$$

Subtracting

The End

Rule 1. No new rules

Subtracting works the same as adding!

With the same denominator:

$$\frac{x}{y} - \frac{z}{y} = \frac{x-z}{y}$$

and with different denominators:

$$\frac{x}{y} - \frac{z}{w} = \frac{x \times w - y \times z}{y \times w} = \frac{xw - yz}{yw}$$

Example:

$$\frac{4}{7} - \frac{2}{7} = \frac{4-2}{7} = \frac{2}{7}$$

Example:

$$\frac{1}{3} - \frac{1}{3} = \frac{0}{3} = 0$$

Example:

$$\frac{3}{4} - \frac{3}{5} = \frac{3 \times 5 - 3 \times 4}{4 \times 5} = \frac{15-12}{20} = \frac{3}{20}$$